SIMPLICIAL RESOLUTION OF THE AUGMENTATION FOR O-MODULES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-simplicial-resolution-augmentation-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof is the author's complete argument: the explicit direct-sum computation plus a parallel-proof delegation to the registered sibling lemma-simplicial-resolution-augmentation (card X1-sp-simplicial-simplicial-resolution-augmentation) with the substitution fully specified in the source text - accepted per the good-sub / criterion-very-ample-bis sister-construction precedents.

Allowed standard prerequisites (editorial).

- g_{n!} on modules. Same chapter, \label{lemma-restriction-module-to-components-site}. Lines 2340-2349: describes g_{n!}(a_n^*G) componentwise as a direct sum.
- Sibling: simplicial resolution of the augmentation (sheaves). Same chapter, \label{lemma-simplicial-resolution-augmentation} (registered card X1-sp-simplicial-simplicial-resolution-augmentation). Lines 2352-2356: the rest of the proof is the sibling's with a_m^{-1}G replaced by a_m^*G.

Source reference. Original lemma 0D72: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1459–1459. Prerequisite pointer invoked at proof line(s) 2340; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D78: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1929–1929. Prerequisite pointer invoked at proof line(s) 2354; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above. For a $\mathcal{O}_\mathcal{D}$-module $\mathcal{G}$ there is an exact complex*

$$\ldots \to g_{2!}(a_2^*\mathcal{G}) \to g_{1!}(a_1^*\mathcal{G}) \to g_{0!}(a_0^*\mathcal{G}) \to a^*\mathcal{G} \to 0$$

of sheaves of $\mathcal{O}$-modules on $\mathcal{C}_{total}$. Here $g_{n!}$ is as in Lemma 0D72.

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Proof. Observe that $a^*\mathcal{G}$ is the $\mathcal{O}$-module on $\mathcal{C}_{total}$ whose restriction to $\mathcal{C}_m$ is the $\mathcal{O}_m$-module $a_m^*\mathcal{G}$. The description of the functors $g_{n!}$ on modules in Lemma 0D72 shows that $g_{n!}(a_n^*\mathcal{G})$ is the $\mathcal{O}$-module on $\mathcal{C}_{total}$ whose restriction to $\mathcal{C}_m$ is the $\mathcal{O}_m$-module

$$\bigoplus\nolimits_{\varphi:[n]\to[m]} f_\varphi^* a_n^*\mathcal{G} = \bigoplus\nolimits_{\varphi:[n]\to[m]} a_m^*\mathcal{G}$$

The rest of the proof is exactly the same as the proof of Lemma 0D78, replacing $a_m^{-1}\mathcal{G}$ by $a_m^*\mathcal{G}$. □